

MongoDB Data Types

1. String

Description: Strings are used to store text data. They are the most commonly used data type in MongoDB.

Use Cases:

- Storing names, descriptions, addresses, etc.

Example:

```
{ "name": "Alice" }
```

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2. Integer

Description: Integers are used to store numerical values. They can be 32-bit or 64-bit depending on your needs.

Use Cases:

- Storing age, count, quantity, etc.

Example:

```
{ "age": 30 }
```

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3. Boolean

Description: Booleans store `true` or `false` values.

Use Cases:

- Flags for user status, feature toggles, etc.

Example:

```
{ "isActive": true }
```

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4. Double

Description: Doubles are used to store floating-point numbers.

Use Cases:

- Storing prices, ratings, measurements, etc.

Example:

```
{ "price": 29.99 }
```

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5. Min/Max Keys

Description: Min/Max keys are used to compare values against the lowest and highest BSON elements.

Use Cases:

- Internal use for sorting and comparisons.

Example:

```
{ "minKey": { "$minKey": 1 } }
```

```
{ "maxKey": { "$maxKey": 1 } }
```

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6. Arrays

Description: Arrays store a list of values or embedded documents.

Use Cases:

- Storing lists of items, tags, embedded documents, etc.

Example:

```
{ "tags": ["mongodb", "database", "NoSQL"] }
```

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7. Timestamp

Description: Timestamps are used to store the date and time.

Use Cases:

- Storing created or modified times for documents.

Example:

```
{ "createdAt": { "$timestamp": { "t": 1621515268, "i": 1 } } }
```

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8. Object

Description: Objects (or embedded documents) are used to store documents within documents.

Use Cases:

- Storing complex data structures and relationships.

Example:

```
{  
  "address": {  
    "street": "123 Main St",  
    "city": "Springfield",  
    "state": "IL"  
  }  
}
```


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9. Null

Description: Null values represent missing or unknown values.

Use Cases:

- Indicating that a field does not have a value.

Example:

```
{ "middleName": null }
```

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10. Symbol

Description: Symbols are similar to strings but are generally reserved for languages that support symbols.

Use Cases:

- Reserved for use by specific languages that support symbols.

Example:

```
{ "symbol": { "$symbol": "symbol_value" } }
```

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11. Date

Description: Dates store the date and time in milliseconds since the Unix epoch.

Use Cases:

- Storing birthdates, event dates, timestamps, etc.

Example:

```
{ "birthdate": { "$date": "1990-01-01T00:00:00Z" } }
```

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12. Object ID

Description: Object IDs are unique identifiers for documents.

Use Cases:

- Used as a unique identifier for documents in a collection.

Example:

```
{ "_id": { "$oid": "60adff3a4e9f5e8a5c8b4567" } }
```

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13. Binary Data

Description: Binary data stores binary data.

Use Cases:

- Storing images, files, and other binary content.

Example:

```
{ "binaryData": { "$binary": { "base64": "binary_data_here", "subType": "00" } } }
```

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14. Code

Description: Code stores JavaScript code.

Use Cases:

- Storing and executing server-side JavaScript code.

Example:

```
{ "code": { "$code": "function() { return 'Hello, world!'; }" } }
```

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15. Regular Expression

Description: Regular expressions store regular expressions for pattern matching.

Use Cases:

- Pattern matching within strings.

Example:

```
{ "pattern": { "$regex": "^MongoDB", "$options": "i" } }
```